Lecture 9

Characteristics and Applications of Polymers

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Week 6 Lectures

In this week's lectures we will study

- Characteristics of polymers
- Structure – property relations
- Polymer processing

Background information:

For this week's lectures read

Callister and Rethwisch 9e Chapter 15 and Chapter 17.13-17.16

QMPlus QXU4006 MS2 – Polymers
QXU4000 MS1 – Polymers

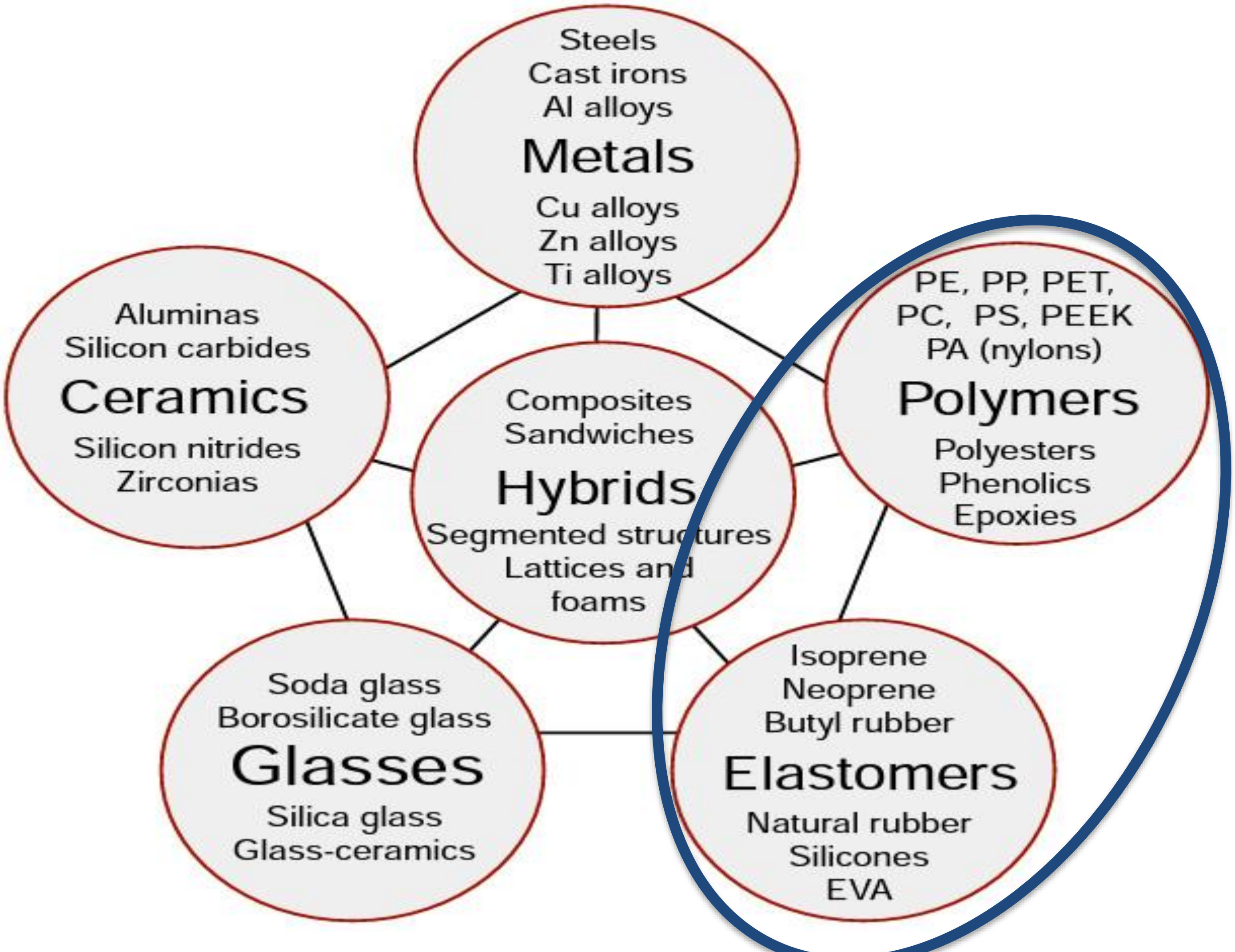
Aims of session & Learning outcomes

- Present the classification and characteristics of polymers
- Discuss polymer structure and its relationship to properties
- Identify the chemistry and structure of polymers
- Describe applications of polymer materials

Chapters 5 & 15 Callister 9e

- Describe characteristics of polymers based on their composition and structure
- Explain the applications of polymers based on their structure
- Understand crystallization and processing of polymers

Materials Families



Materials: Engineering, Science, Processing and Design by Ashby, Shercliff and Cebon. 2007 ISBN-13: 978-0-7506-8391-3 pg 15

Classification of polymers

How can we classify polymers?

- **Thermoplastic** - held together by secondary interactions
  soften on heating, harden on cooling
  – can be repeated many times (reversible)

- **Thermoset** - held together by primary interactions
  harden permanently
  – decompose on heating

Classification of polymers

Or by their behaviour characteristics?

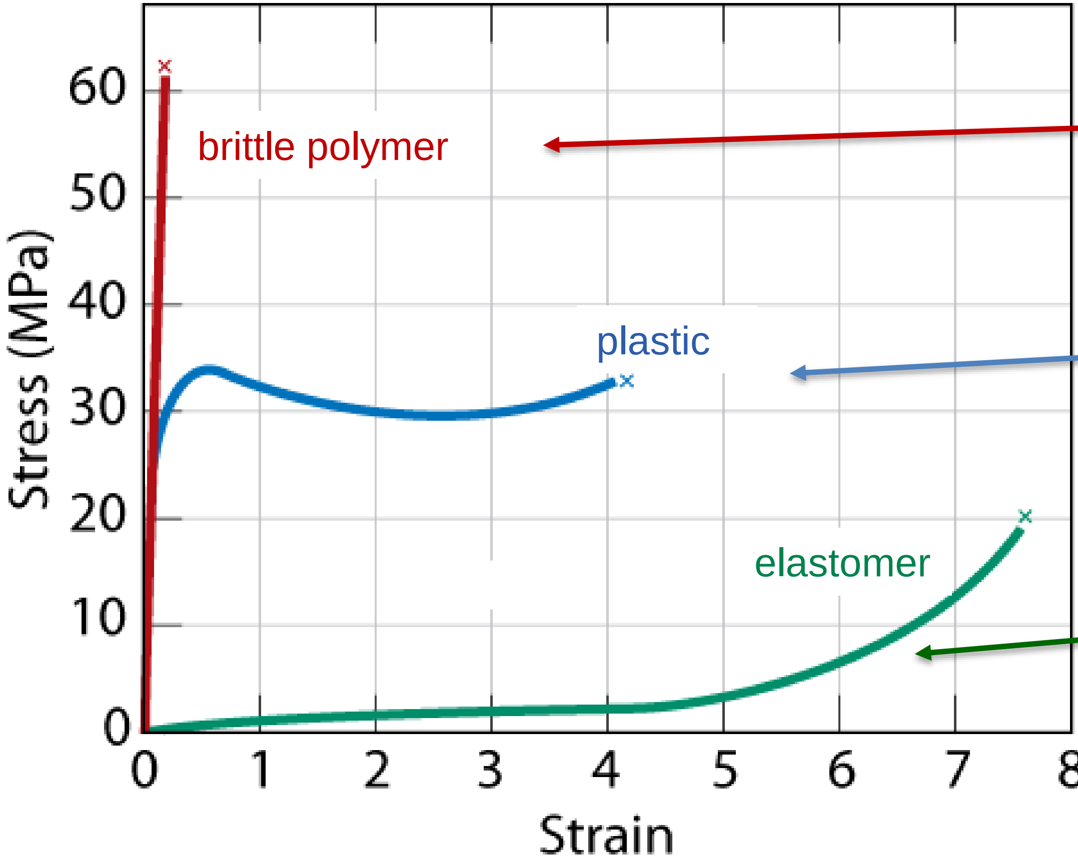
- **Glassy** – brittle
  amorphous – frozen (immobile)

- **Rubbery** – plastic
  amorphous or semi-crystalline (mobile)

- **Elastomeric** – extendible
  amorphous or semi-crystalline (constrained by x-links)

Classification of polymers

Or by their behaviour characteristics?



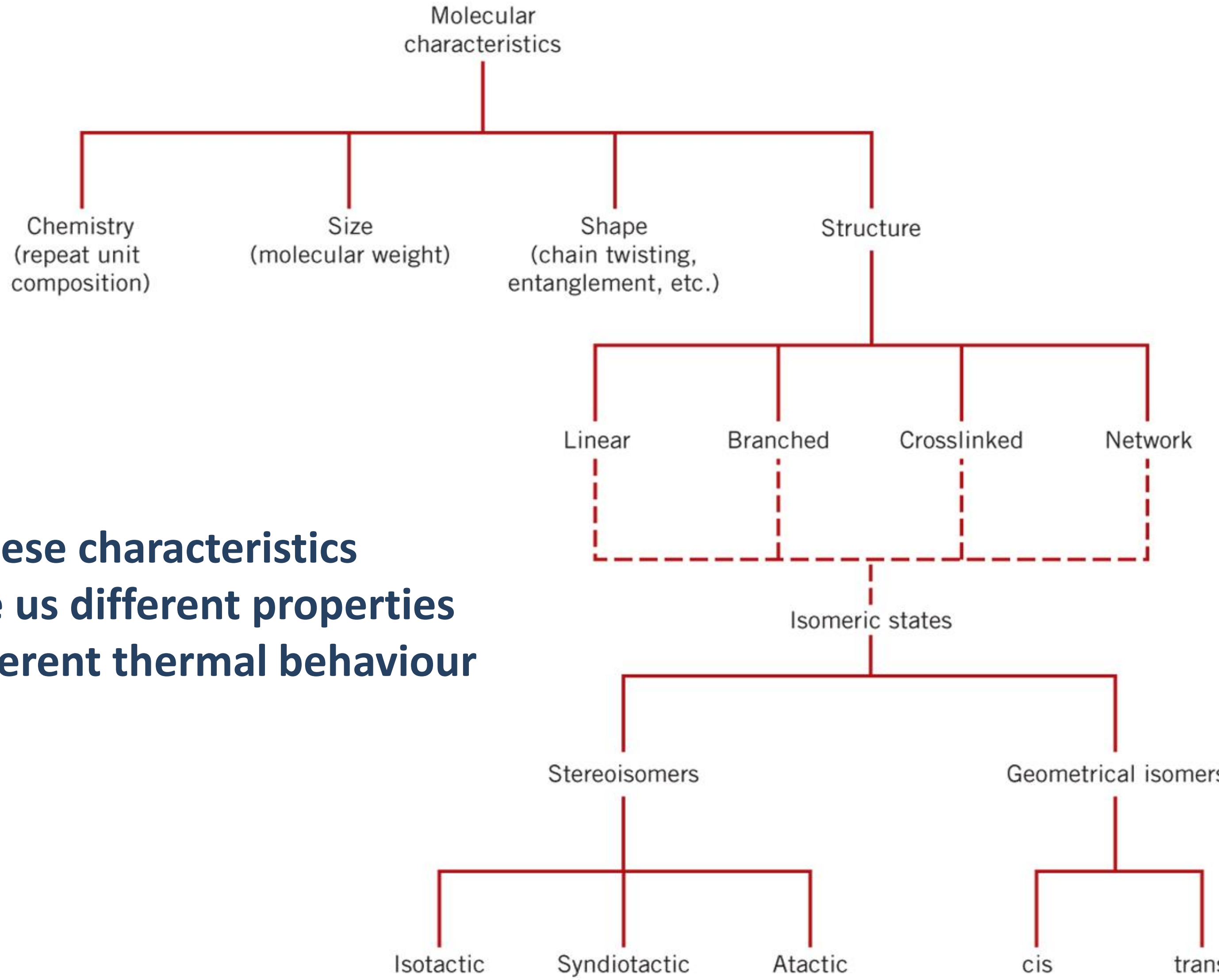
Low strain to failure

Large plastic strain

Large elastic strain

Fig 15.1 Callister & Rethwisch 9e

Polymer characteristics



All of these characteristics can give us different properties and different thermal behaviour

Fig 5.8 Callister & Rethwisch 9e

Thermal behaviour

3 important considerations for **polymer processing**:

- **Melting** – change to liquid state
- **Glass transition** – change in chain mobility
- **Crystallization** – regions of regular order

Related to:

- Degree of order / disorder
- Free volume

Need to understand:

How these change with polymer characteristics

How these change with temperature

Thermal behaviour

Melting:

- Melting temperature, T_m
- Transformation from solid to liquid

- Melting takes place over a range of temperature (different from metals and ceramics)

- T_m Depends on:
  - Mw distribution
  - Rate of heating
  - Degree of crystallization (so previous thermal history)

Thermal behaviour

Glass transition:

- Amorphous and semi-crystalline polymers
- Change from rigid to rubbery state
- Disorder
- Free volume change

Transition to rubbery when there is enough space (free volume) for polymer chains to reconfigure and change shape

– by rotation of the polymer backbone

Increasing temperature, increasing plasticizer

Thermal behaviour

Glass transition:

- Free volume change at T_g
- Less for semi-crystalline
- No T_g for highly crystalline or cross-linked polymers

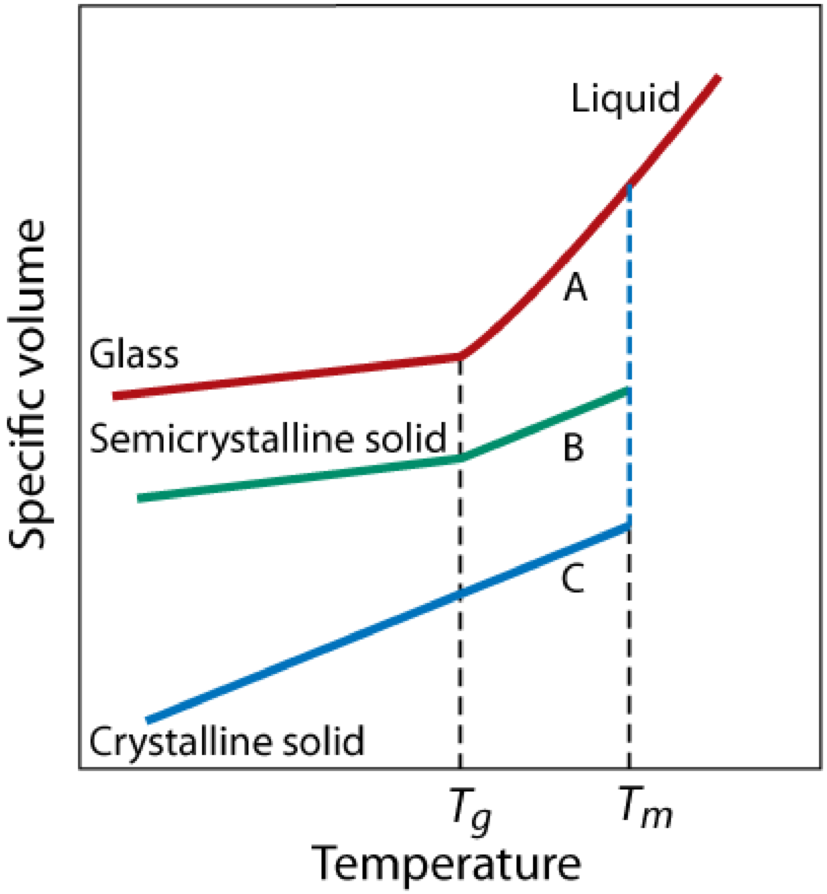


Fig 15.18 Callister & Rethwisch 9e

Thermal behaviour

Glass transition:

- Other changes that occur with glass transition:

- Stiffness of the polymer
- Heat capacity
- Coefficient of thermal expansion

All associated with interactions between the polymer chains

Thermal behaviour

Changes in T_g and T_m with crystallization and M_w

Dependence of polymer properties and melting and glass transition temperatures on molecular weight

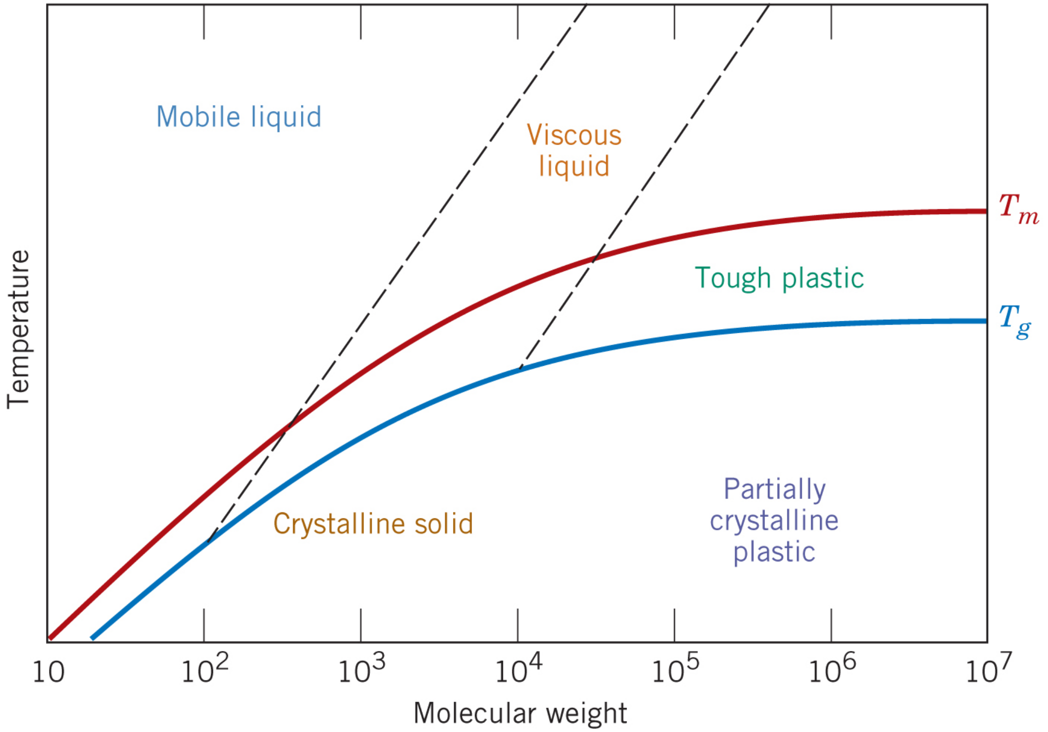


Fig 15.19 Callister & Rethwisch 9e

Thermal behaviour

What factors affect T_m and T_g?

- Both T_m *and* T_g increase with increasing **chain stiffness**

- Chain stiffness increased by presence of
  1. Bulky sidegroups
  2. Polar groups or sidegroups
  3. Chain double bonds and aromatic chain groups

- Regularity of repeat unit arrangements (tacticity)
  – affects T_m only

For homopolymers $T_g \sim 0.5T_m - 0.8T_m$ (Kelvin temperature)
(Can change this for blends and co-polymers)

Thermal behaviour

Ranges of T_g and T_m for different polymers

Material	*Glass Transition Temperature (°C)*	*Melting Temperature (°C)*
Polyethylene (low density)	−110	115
Polytetrafluoroethylene	−97	327
Polyethylene (high density)	−90	137
Polypropylene	−18	175
Nylon 6,6	57	265
Poly(ethylene terephthalate) (PET)	69	265
Poly(vinyl chloride)	87	212
Polystyrene	100	240
Polycarbonate	150	265

Table 15.2 Callister & Rethwisch 9e

Thermal behaviour

Changes with crystallization

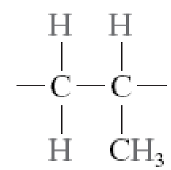
- T_m and T_g depend on:

  - Molecular weight distribution - range of *Tm*
  - Chain length - long chains = higher T_m
  - Degree of branching - more space = lower T_g
  - Bulky side groups - less movement = higher T_g
  - Polar groups - stronger interactions = higher T_g
  - Double bonds - stiffer chains = higher T_g
  - Degree of crosslinking - higher T_g (or no T_g at all)

Thermal behaviour

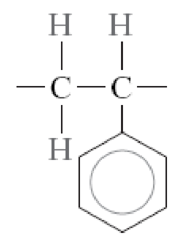
Examples:

Polypropylene T_g = -18°C



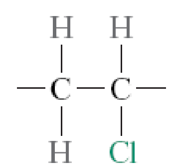
Non-polar, small side group

Polystyrene T_g = 100°C



Large ‘bulky’ side group

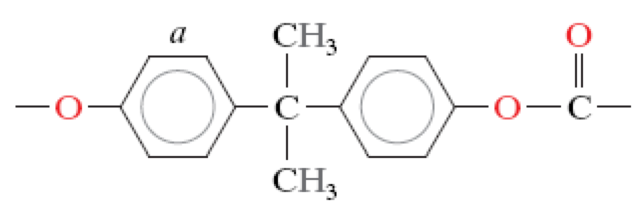
Polyvinylchroride T_g = 87 °C



Polar side group

Double bonds and aromatic rings **stiffen chains**

e.g. Polycarbonate T_g = 150 °C



Summary

There are many ways to classify polymers:

By their:

- Chemistry (repeat units)
- M_w and structure / crystallinity
- Thermal behaviour (T_g, T_m)
- Mechanical behaviour – glassy, rubbery, elastomer

All of these are important in their processing